

Hyper PLA Filament Technical Data Sheet

Version 1.0

1. Product introduction

Hyper PLA filament is a 3D printing filament developed based on PLA. It has good fluidity and fast curing characteristics. It is compatible with high-speed and low-speed printing, supports printing speeds up to 600 mm/s, and has high mechanical properties and tensile strength. Compared with ordinary ABS filaments, Hyper PLA has a low shrinkage rate, no edge warping during printing, and printing is successful.

2. Physical Performance Parameters

Items	Testing Criteria	Parameters
Density	ASTM D792 (ISO 1183, GB/T 1033)	1.24g/cm ³
Melt index	190°C, 2.16kg	7.15g/10min
Glass transition temperature	DSC, 10°C/min	60.3°C
Melting temperature	DSC, 10°C/min	168.7°C
Vicat softening temp.	ASTM D1525 (ISO 306, GB/T 1633)	59.0°C
Heat deflection temp.	ASTM D648 (ISO 75-2, GB/T 1634)	55.5°C (0.45Mpa)
		54.3°C (1.80Mpa)

3. Mechanical Properties

Categories	Items	Testing Criteria	Parameters
Tensile strength	Tensile strength (X-Y)	ASTM D638 (ISO 527, GB/T 1040)	49MPa
	Tensile modulus (X-Y)	ASTM D638 (ISO 527, GB/T 1040)	2445MPa
	Elongation at break (X-Y)	ASTM D638 (ISO 527, GB/T 1040)	14.3%
	Tensile strength (Z)	ASTM D638 (ISO 527, GB/T 1040)	23MPa
	Tensile modulus (Z)	ASTM D638 (ISO 527, GB/T 1040)	1982MPa
	Elongation at break (Z)	ASTM D638 (ISO 527, GB/T 1040)	1.7%
Bending strength	Bending strength (X-Y)	ASTM D790 (ISO 178, GB/T 9341)	80MPa
	Bending modulus (X-Y)	ASTM D790 (ISO 178, GB/T 9341)	2863MPa
Bending strength	Bending strength (Z)	ASTM D790 (ISO 178, GB/T 9341)	45MPa
	Bending modulus (Z)	ASTM D790 (ISO 178, GB/T 9341)	2493MPa

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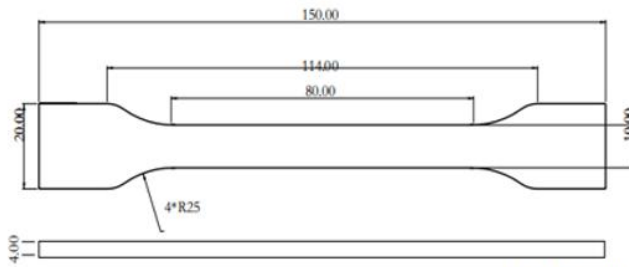
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Categories	Items	Testing Criteria	Parameters
Charpy impact strength	Charpy impact strength (X-Y)	ASTM D256 (ISO 179, GB/T 1043)	50.2kJ/m ² 13.1kJ/m ² (notched)
	Charpy impact strength (Z)	ASTM D256 (ISO 179, GB/T 1043)	4.9kJ/m ²

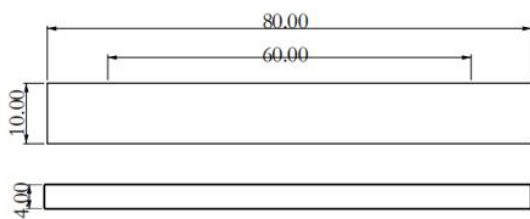
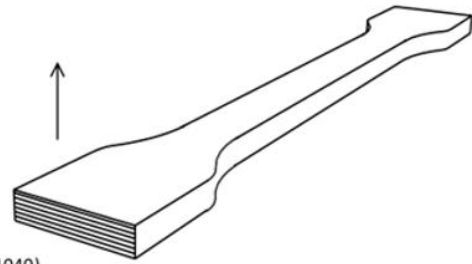
Printing parameters and styles of printing conditions:

Print Conditions	Parameters
Nozzle Temperature	220℃
Hot Bed Temperature	50℃
Chamber temperature	35℃
Fill style, Packed density	Concentric circle, 100%
Cooling fan	Open
Annealing or not annealing	NO



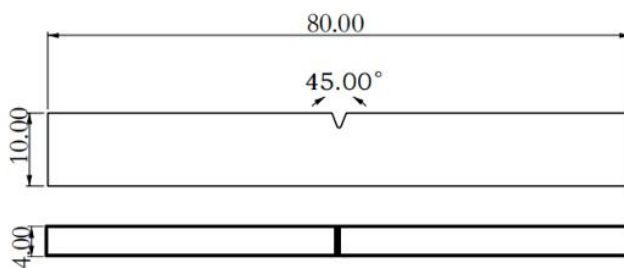
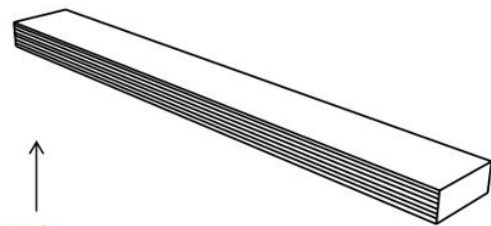
ASTM D638 (ISO 527, GB/T 1040)

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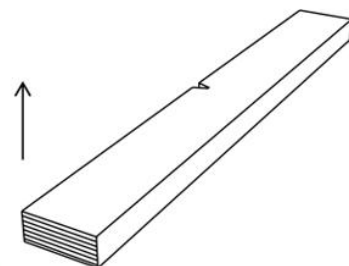
ASTMD790 (ISO 178, GB/T 9341)

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ASTM D256 (ISO 179, GB/T 1043)

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4. Recommended printing conditions

Print Temperature	Hotbed Temperature	Chamber Temperature	Print Speed	Cooling fan	Pumping Distance	Pumping speed
190-240℃	45-60℃	10-35℃	30-240mm/s	100%	0.4-1.8mm	20-60mm/s

5. Other parameter conditions

Drying temp. and time	Heated bed plate	Suitable nozzle size	Annealing temp. and time	Supporting materials
50-60℃, 8-12h	PEI Platform plate	0.2/0.4/0.6/0.8	Not necessary	Support for PLA, BVOH and PVA

6. Compatible Models

Hyper PLA is widely used in FDM 3D printers on the market and is compatible with both high-speed and conventional-speed printers.

7. Storage Condition

Please place this product in a dry and ventilated environment, avoiding high temperatures, direct sunlight, or humid conditions. If it is not used up shortly after opening, it is recommended to store it in a drying box for future use.

8. Disclaimer

The values given in this data sheet are for reference and comparison only. Actual values may vary with printing conditions, and the end-use performance of printed models depends on model designs, environmental conditions, printing conditions, etc.